



**RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES**

**Bachelor of Science in Applied Sciences
First Year - Semester I Examination – February 2024**

COM 1201 – PRINCIPLES OF PROGRAM DESIGN

Time: Two(02) hours

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- **Answer ALL the questions.**
 - **This paper contains four(04) questions in three (03) pages**
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1. a) What is an algorithm? Discuss three (03) features of a good algorithm. (05 marks)
- b) State and define two (02) methods of representing algorithms. (04 marks)
- c) Compare the differences between machine Language and computer programming language. (06 marks)
- d) What are the differences between a compiler and an interpreter? List two (02) programming languages examples for each. (06 marks)
- e) What is the difference between syntax errors and semantic errors? (04 marks)

2. a) List four (04) advantages of using flow charts. (04 marks)
- b) Draw and briefly explain five (05) symbols commonly used in flowcharts. (05 marks)

- c) ABC hospital allows its registered customers to access their lab reports online through an internet-portal on or after the specified date. A unique reference number is given for each lab test and is printed on the payment receipt. The lab report will be ready for online view on or after the specified date printed on the receipt. To access the report, the customer has to input the reference number, if the entered reference is validated successfully, the system sends the test report to the customer's email address.

After sending the email, the system will display the message "Please check your email."

If the customer tries to access the report before the specified date, the system displays the message "Your report will be available on the date: <available date>"

The reference number is valid only for fifteen days after the date on which the report becomes available.

Draw a flowchart to show the above process from customer sign-in to display message.

(06 marks)

- d) The ATM machines of ABC Bank use two-factor authentication via an SMS besides conventional PIN control. The customers of ABC bank have a choice either to use a card and a PIN or an SMS authentication. Other banks customers can also withdraw cash from ATMs of ABC bank but only with a Card and a PIN option. After three consecutive unsuccessful attempts, a customer's card is seized by the machine.

Draw a flowchart only for the cash withdrawal process of ABC's ATM.

(10 marks)

3. a) Discuss two (02) types of Repetition Structures which can be use in pseudocode design.

(06 marks)

- b) Define a pseudocode function called "calculateArea" that takes parameters length and width and returns the area of a rectangle

(06 marks)

- c) Write a pseudo algorithm to find the sum of all even numbers between 1 and 100 inclusive.

(06 marks)

- d) Write the expected outputs of following pseudocodes

- i. Start
Use variable : average of type float
Get the sum
 $\text{average} = \text{sum} / 6$
Output the average
Stop

(03 marks)

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ii.    valid_input = false
        While not valid_input Do
            Print "Please enter a number:"
            Input number
            IF number IS Numeric Then
                valid_input = true
            Else
                Print "Invalid input. Please enter a numeric value."
            Endif
        Endwhile

```

(04 marks)

4. a) Define what a programming paradigm is and discuss why understanding different paradigms is important for a software developer. (06 marks)

b) Compare and contrast the procedural and object-oriented programming paradigms. Provide examples of languages that represent each paradigm. (06 marks)

c) Explain the concept of concurrency in the context of programming paradigms. How do different paradigms handle concurrent programming? (05 marks)

d) Briefly discuss following terms.

i. Imperative programming paradigm (04 marks)

ii. Declarative programming paradigm (04 marks)

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